

Social Media Engagement Analytics Using Python

Project Summary

Source Dataset:

https://github.com/GeethaGunasekaran1/Dataset_rep/blob/main/social_media_engagement_5000.csv

1. Project Overview

This project analyzes social-media engagement data using Python. The workflow covers data import, data cleaning, transformation, exploratory analysis, feature engineering, statistical analysis, and visualization using Pandas, NumPy, Matplotlib, Seaborn, and Plotly.

2. Dataset Overview

Item	Details
Rows	5,000
Columns	19
Countries	10
Post types	4
Content categories	8
Devices	3
Sentiments	3

Main column groups:

- User: user_id, age, gender, country, follower_count, is_verified.
- Post: post_id, post_type, post_category, posted_at, hashtags.
- Engagement: likes, comments, shares, watch_time_sec, impression_count, engagement_rate.
- Context: device_type and sentiment.

3. Python Libraries

- NumPy – numerical calculations and array operations.
- Pandas – data loading, cleaning, transformation, grouping and analysis.
- Matplotlib – static charts and basic visualization.
- Seaborn – statistical and categorical visualizations.
- Plotly Express – interactive visualizations.

4. Data Import & Setup

- The CSV dataset is loaded into a Pandas DataFrame.
- The posted_at column is converted to datetime using dayfirst=True.
- Initial inspection uses head(), tail(), shape, columns, info(), dtypes and describe().

5. Data Cleaning & Transformation

- Missing-value analysis found 150 missing values each in age, gender, likes, comments, shares and sentiment.
- The dataset contains 0 duplicate rows.

- Text/category fields are stripped, converted to string and standardized to lowercase.
- is_verified is standardized as a Boolean field.
- Numeric columns are converted to numeric data types.
- Negative values in selected numeric fields are converted to missing values.
- Ages below 13 or above 100 are treated as missing.
- Missing age, likes, comments and shares are filled using the median.
- Missing gender and sentiment are filled using the mode.
- Sentiment is restricted to positive, neutral and negative.

6. Feature Engineering

- hashtag_count is created by counting hashtag tokens beginning with #.
- engagement_score is calculated as likes + comments + shares.
- Date features are derived from posted_at, including year, month, month_name and day_of_week.
- A calculated engagement-rate check is created using $\text{engagement_score} / \text{impression_count} \times 100$.

7. Exploratory Data Analysis

- value_counts() and nunique() are used to understand categorical distributions.
- Correlation analysis is performed for age, engagement metrics, impressions, followers, hashtag_count and engagement_score.
- groupby() is used to compare performance across post types and countries.
- Grouped summaries are also created for post category and sentiment.

8. Statistical Analysis

Metric	Mean	Median	Std. Dev.	P90
Likes	10,107.00	10,105.50	5,702.29	18,016.30
Comments	1,502.04	1,497.00	856.39	2,695.10
Shares	1,002.91	1,012.00	570.86	1,795.10
Watch time (sec)	4,014.50	4,034.50	2,308.10	7,212.10
Engagement rate	0.9644	0.2539	5.3180	1.2012
Follower count	393,698.22	388,982.00	230,927.88	717,755.90

The engagement-rate distribution is strongly right-skewed (skewness ≈ 18.78) and has very high kurtosis (≈ 482.99), indicating extreme values in the distribution.

9. Visualization

Library	Visualizations	Purpose
Matplotlib	Scatter, line, bar, pie, histogram, box plot	Relationships, trends and distributions
Seaborn	Count plot, bar plot, violin, pair plot, heatmap, swarm plot	Category comparisons and statistical patterns
Plotly	Interactive line and bubble charts	Interactive performance analysis

10. Key Results

Dimension	Highest average engagement rate	Rate
Post type	video	1.1224

Category	food	1.3586
Sentiment	negative	1.0385
Country	brazil	1.5407

11. Post Type Results

- Average likes: video 10,188.60; image 10,104.87; text 10,100.15; reel 10,037.80.
- Average engagement rate: video 1.1224; text 1.0645; image 0.8956; reel 0.7830.

12. Category Results

- Food has the highest average engagement rate in the displayed category summary at 1.3586.
- Tech follows at 1.1605 and lifestyle at 1.0910.
- Travel has the lowest displayed category engagement rate at 0.7022.

13. Country Results

- Brazil has the highest displayed average engagement rate at 1.5407.
- Australia: 1.3243; France: 1.1464; UAE: 1.1124; Canada: 0.9167.
- UK: 0.8510; Japan: 0.7696; Germany: 0.7590; India: 0.6550; USA: 0.5766.

14. Final Output from the Notebook

The notebook's final insight cell prints image as the highest-engagement post type, education as the best-performing category, Australia as the highest-engagement country, and negative sentiment as best-performing.

Note: The grouped tables earlier in the notebook show different leaders for post type, category and country (video, food and Brazil). Both outputs are retained here exactly as represented in the source notebook.

15. Overall Workflow

Import → Inspect → Clean → Transform → Explore → Group & Aggregate → Feature Engineering → Statistical Analysis → Visualize → Extract Insights